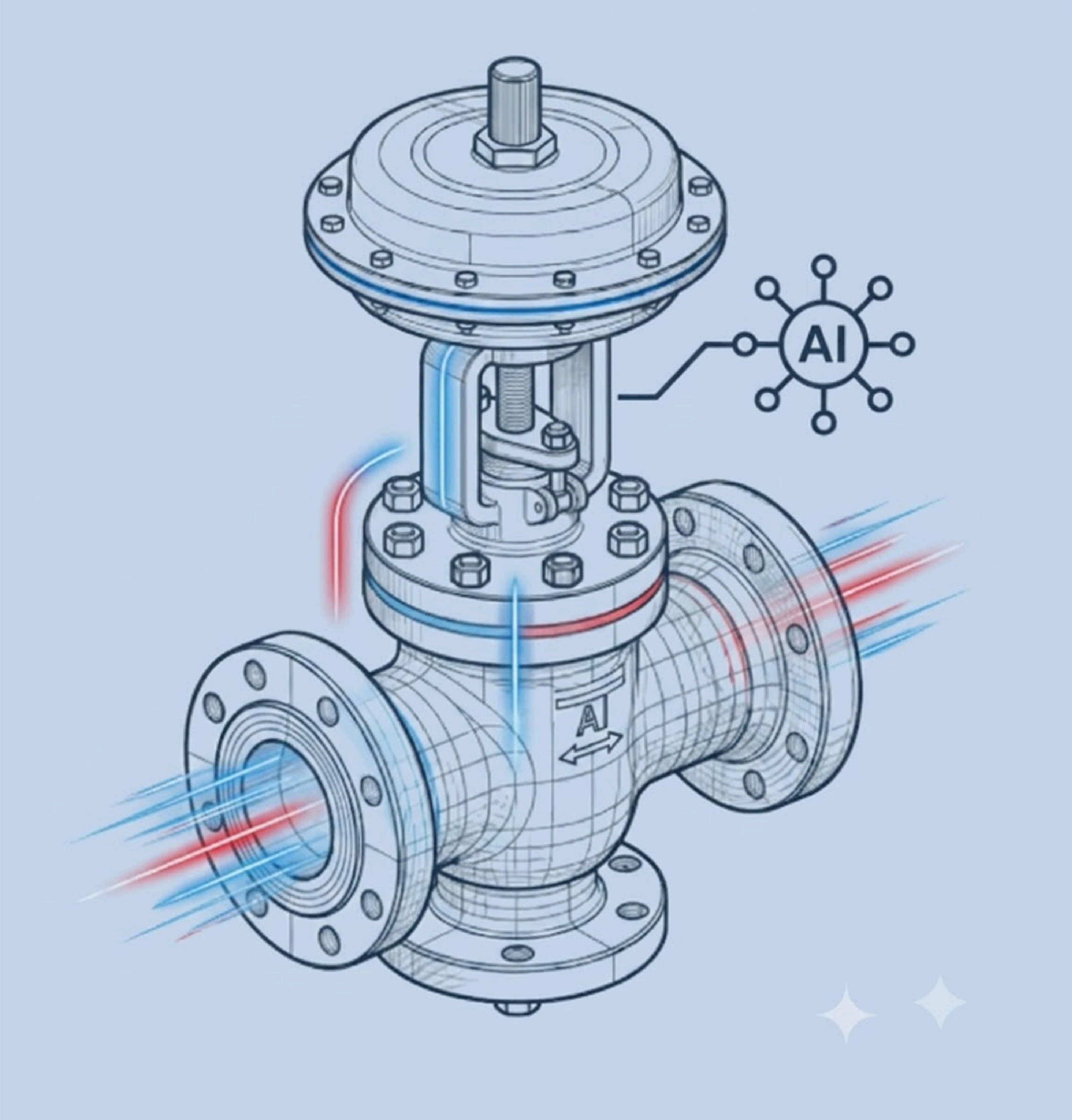
TITLE PAGE

- **Problem Statement ID** – Q3
- **Problem Statement Title** – Autonomous Production Choke Controller
- **Theme** – Industrial Automation & AI
- **PS Category** – Software
- **Student Name** – Arpita Nanda
- **Student ID** – 23BCE11660



Autonomous Digital Twin & Lagrangian MPC Controller

Detailed explanation of the proposed solution:

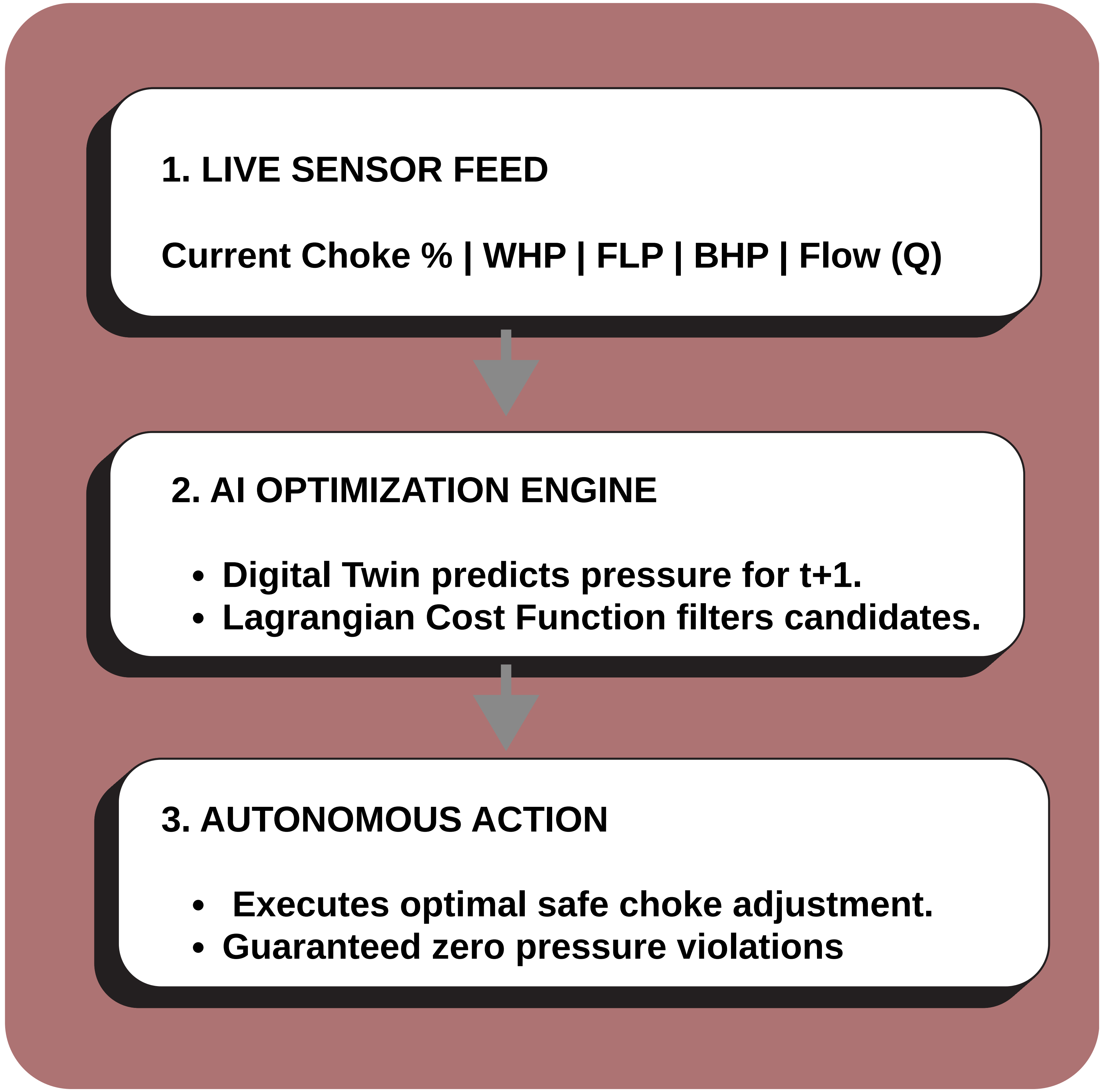
- An automated AI controller for oil well chokes, replacing manual heuristic adjustments.
- ML Digital Twin: Predicts next-step (t+1) pressure responses based on current well states.
- MPC Optimizer: Evaluates hundreds of valid choke moves (within +-5% rate limits) in real-time.

How it addresses the problem:

- Maximizes Output: Dynamically drives the well to reach and sustain target oil production.
- Absolute Safety: Strictly maintains Wellhead (WHP), Flowline (FLP), and Bottom-Hole (BHP) pressures within physical limits.

Innovation and uniqueness of the solution:

- Lagrangian Penalties: Applies massive mathematical costs (+10,000) to unsafe moves, forcing the AI to select only safe actions.
- Zero-Overshoot: Autonomously rejects unachievable target requests before they are executed.



TECHNICAL APPROACH

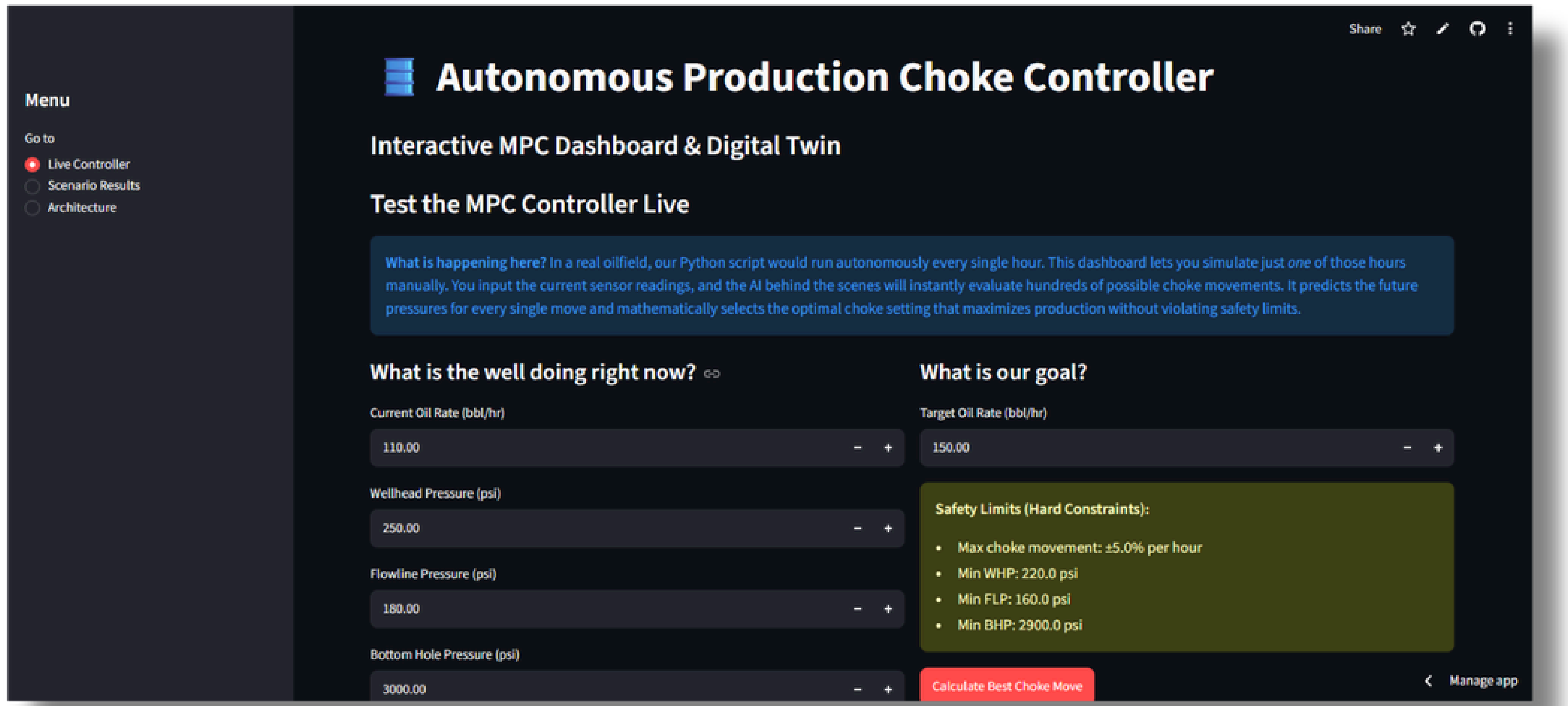
Technologies Used:

- **Core Logic**: Python 3.11+, scikit-learn (Random Forest Regressor), pandas, numpy.
- **Control Architecture**: Model Predictive Control (MPC) with Lagrangian mathematical penalty functions.
- **Cloud & Infrastructure:**
  - AWS EC2 / S3: Enterprise-grade compute and telemetry data storage.
  - AWS Bedrock: (Future scope integration for GenAI-driven anomaly insights).
- **Deployment**: Streamlit Cloud for real-time interactive UI and remote dashboard access.

Methodology & Implementation Process:

- **Step 1**: Telemetry Ingestion: Live wellhead data (WHP, FLP, BHP, Q) is streamed into the cloud environment.
- **Step 2**: Digital Twin Simulation: The ML model rapidly simulates pressure responses for all valid choke movements ($\pm 5\%$ rate limits).
- **Step 3**: Lagrangian Filtering: The cost function mathematically eliminates any trajectory that breaches safety constraints (+10,000 penalty).
- **Step 4**: Autonomous Execution: The absolute optimal, zero-violation choke position is pushed to the live Streamlit dashboard for operator execution.

Dashboard Visual



SCAN FOR LIVE DEMO & SOURCE CODE

Live App

GitHub

FEASIBILITY AND VIABILITY

- **High Commercial Viability**: The solution leverages existing open-source frameworks (Python, Scikit-learn) and standard cloud infrastructure (AWS), requiring zero proprietary hardware investments.
- **Computational Efficiency**: The Lagrangian MPC algorithm is mathematically lightweight, allowing it to run seamlessly on standard edge or cloud compute environments.
- **Rapid Integration**: Can be deployed as a modular microservice directly into an operator's existing SCADA or control systems.

- **Risk 1**: Concept Data Drift: Physical oil reservoir dynamics naturally shift over time, which could slowly degrade the ML Digital Twin’s prediction accuracy.
- **Risk 2**: Network Latency: Relying strictly on cloud servers (AWS) for processing could introduce micro-delays in mission-critical, real-time choke adjustments if connectivity drops.

- **Mitigating Drift** (MLOps): Implement an automated CI/CD retraining pipeline that continuously updates the Digital Twin with the latest well telemetry data to ensure peak accuracy.
- **Mitigating Latency** (Edge Deployment): Transition the final MPC execution module from the cloud directly to Edge-Compute Nodes (industrial IoT devices) located physically at the wellpad for zero-latency control.

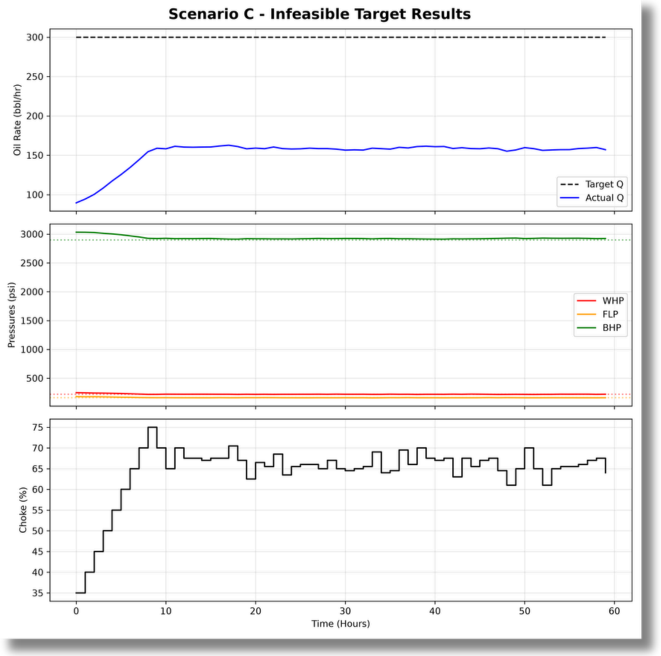
ARTIFACTS

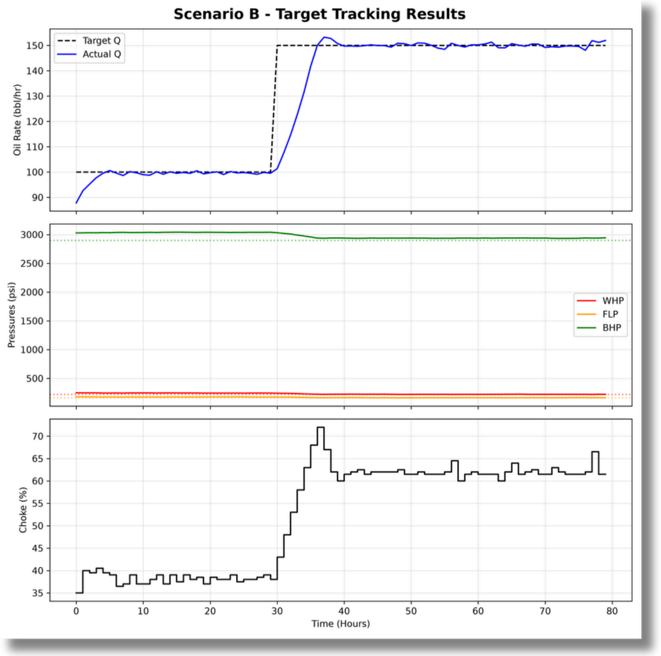
- **Copy of the Code Embedded**: The core Python optimization engine (left) featuring the Lagrangian MPC cost function. This demonstrates the mathematical penalties used to guarantee absolute safety and zero-overshoot.
- **Dashboard Snaps**: Live Streamlit UI visualizations (left) displaying real-time monitoring of Wellhead (WHP) and Bottom-Hole (BHP) pressures against physical safety limits.
- **Snaps of the Solution Proposal**: Interactive telemetry graphs proving the AI controller's ability to safely and dynamically adjust the choke to reach the target oil flow rate without breaching thresholds.

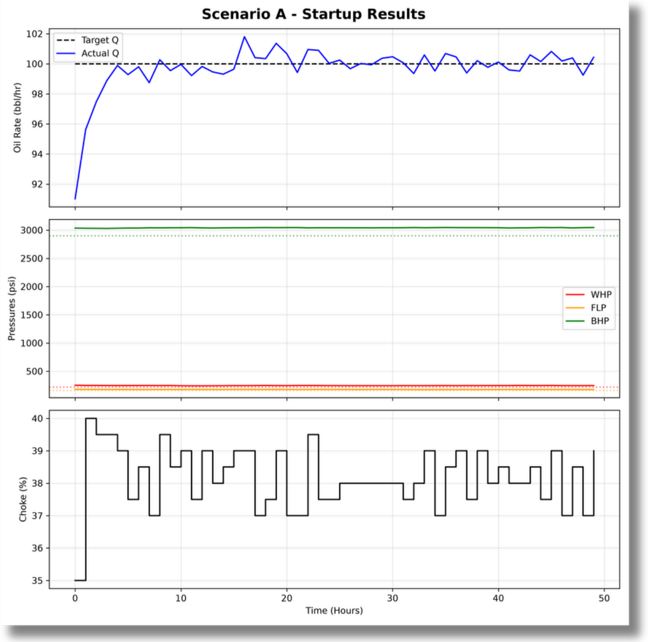
LAGRANGIAN MPC PENALTY FUNCTION

```python
        # LAGRANGIAN OBJECTIVE COST FUNCTION
        # Base Objective: Minimize error between target and predicted flow rate
        base_cost = abs(target_q - q1[0])

        # Penalty Weights: Massive mathematical penalty for constraint violations
        penalty = 0.0
        if whp1[0] < self.hard_min_whp: penalty += (self.hard_min_whp - whp1[0]) * 10000
        if flp1[0] < self.hard_min_flp: penalty += (self.hard_min_flp - flp1[0]) * 10000
        if bhp1[0] < self.hard_min_bhp: penalty += (self.hard_min_bhp - bhp1[0]) * 10000

        total_cost = base_cost + penalty

        # The optimizer searches for the candidate with the lowest possible cost
        if total_cost < lowest_cost:
            lowest_cost = total_cost
            best_candidate = candidate
            self.temp_prediction = {'q': q1[0], 'whp': whp1[0], 'flp': flp1[0], 'bhp': bhp1[0]}

    self.last_prediction = self.temp_prediction
    return best_candidate
```

REAL-TIME PRESSURE CONSTRAINTS







RESEARCH AND REFERENCES

1. Project Artifacts & Live Deployment

- **Complete Source Code** (GitHub):https://github.com/Arpitananda123/Honeywell-Choke-Controller
- **Live Working Prototype** (Streamlit):https://honeywell-choke-controller-wdhuqcjxdofxc4davwekfk.streamlit.app/

2. Technical Frameworks & Documentation

- **Scikit-Learn** (Digital Twin ML):https://scikit-learn.org/stable/modules/ensemble.html
- **Streamlit Cloud** (UI/UX):https://docs.streamlit.io/
- **AWS Infrastructure** (Telemetry):https://aws.amazon.com/architecture/

3. Academic & Industry Literature

- **Society of Petroleum Engineers** (SPE): Research on Model Predictive Control (MPC) for autonomous well optimization (https://onepetro.org/)
- **IEEE Xplore**: Applied Lagrangian penalty optimization in industrial closed-loop AI systems (https://ieeexplore.ieee.org/)